Identifying Data Requirements

Online Shopping Platform (Amazon / Flipkart style) - DBMS Project

Objective

To identify all the data that must be stored in the database of an online shopping platform. Below are the entities, their attributes and their primary keys (PK). Foreign keys (FK) are shown only to indicate how the entities are linked.

Summary of Entities and Primary Keys

Entity	Primary Key	Purpose
Customer	customer_id	A person who registers on the platform and places orders.
Category	category_id	A group under which products are organised (e.g., Electronics, Books).
Supplier	supplier_id	A seller or vendor who provides products to the platform.
Product	product_id	An item listed for sale on the platform.
Order	order_id	A purchase request placed by a customer.
Payment	payment_id	The payment made for an order.
Shipment	shipment_id	The delivery of an order to the customer.
Review	review_id	A rating and comment given by a customer for a product.

Entities with Attributes

1. Customer

A person who registers on the platform and places orders. **Primary key: customer_id**

Attribute	Data Type	Key	Description
customer_id	INT	**PK**	Unique ID for each customer
first_name	VARCHAR(50)		Customer's first name
last_name	VARCHAR(50)		Customer's last name
email	VARCHAR(100)		Login email (unique)
password_hash	VARCHAR(255)		Encrypted password
phone	VARCHAR(15)		Contact number
address	VARCHAR(255)		Default delivery address
city / state / pincode	VARCHAR		Address details
registration_date	DATE		Date the account was created

2. Category

A group under which products are organised (e.g., Electronics, Books). **Primary key: category_id**

Attribute	Data Type	Key	Description
category_id	INT	**PK**	Unique ID for each category
category_name	VARCHAR(100)		Name of the category
description	VARCHAR(255)		Short description
parent_category_id	INT	**FK**	Optional parent category (sub-categories)

3. Supplier

A seller or vendor who provides products to the platform. **Primary key: supplier_id**

Attribute	Data Type	Key	Description
supplier_id	INT	**PK**	Unique ID for each supplier
supplier_name	VARCHAR(100)		Name of the supplier / company
contact_person	VARCHAR(100)		Person to contact
email	VARCHAR(100)		Supplier email
phone	VARCHAR(15)		Supplier phone
address	VARCHAR(255)		Supplier address

4. Product

An item listed for sale on the platform. **Primary key: product_id**

Attribute	Data Type	Key	Description
product_id	INT	**PK**	Unique ID for each product
product_name	VARCHAR(150)		Name of the product
description	TEXT		Detailed description
price	DECIMAL(10,2)		Selling price
stock_quantity	INT		Units available in stock
category_id	INT	**FK**	Refers to Category
supplier_id	INT	**FK**	Refers to Supplier

5. Order

A purchase request placed by a customer. **Primary key: order_id**

Attribute	Data Type	Key	Description
order_id	INT	**PK**	Unique ID for each order
customer_id	INT	**FK**	Refers to Customer who placed it
order_date	DATETIME		Date and time of the order
order_status	VARCHAR(20)		Placed / Confirmed / Shipped / Delivered / Cancelled
total_amount	DECIMAL(10,2)		Total bill amount
shipping_address	VARCHAR(255)		Address where it must be delivered

6. Payment

The payment made for an order. **Primary key: payment_id**

Attribute	Data Type	Key	Description
payment_id	INT	**PK**	Unique ID for each payment
order_id	INT	**FK**	Refers to Order being paid for
payment_date	DATETIME		Date and time of payment
payment_method	VARCHAR(30)		Card / UPI / Net Banking / COD
amount	DECIMAL(10,2)		Amount paid
payment_status	VARCHAR(20)		Success / Failed / Pending / Refunded
transaction_ref	VARCHAR(50)		Gateway transaction reference

7. Shipment

The delivery of an order to the customer. **Primary key: shipment_id**

Attribute	Data Type	Key	Description
shipment_id	INT	**PK**	Unique ID for each shipment
order_id	INT	**FK**	Refers to Order being shipped
shipped_date	DATE		Date the package left the warehouse
expected_delivery_date	DATE		Estimated delivery date
delivered_date	DATE		Actual delivery date
carrier_name	VARCHAR(50)		Courier / delivery partner
tracking_number	VARCHAR(50)		Tracking ID for the customer
shipment_status	VARCHAR(20)		In Transit / Out for Delivery / Delivered

8. Review

A rating and comment given by a customer for a product. **Primary key: review_id**

Attribute	Data Type	Key	Description
review_id	INT	**PK**	Unique ID for each review
customer_id	INT	**FK**	Refers to Customer who wrote it
product_id	INT	**FK**	Refers to Product being reviewed
rating	INT		Rating from 1 to 5
comment	TEXT		Written feedback
review_date	DATE		Date of the review

Relationships Between Entities

Entities	Relationship
Customer - Order	One customer can place many orders (1 : N)
Order - Payment	One order has one payment record (1 : 1)
Order - Shipment	One order is delivered through one shipment (1 : 1)
Category - Product	One category contains many products (1 : N)
Supplier - Product	One supplier supplies many products (1 : N)

Customer - Review	One customer can write many reviews (1 : N)
Product - Review	One product can receive many reviews (1 : N)
Order - Product	Many-to-many (M : N): an order has many products and a product appears in many orders

Additional Supporting Entity: Order_Item

Because Order and Product have a many-to-many relationship, an extra table **Order_Item** is needed to store which products are in which order. Primary key: **order_item_id** (or composite key order_id + product_id).

Attribute	Data Type	Key	Description
order_item_id	INT	**PK**	Unique ID for each order line
order_id	INT	**FK**	Refers to Order
product_id	INT	**FK**	Refers to Product
quantity	INT		Number of units ordered
unit_price	DECIMAL(10,2)		Price per unit at the time of order